

“CÔNG NGHỆ CHIẾN LƯỢC: VI MẠCH x QUANTUM”



Quantum computing: overview, training, and research

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- PhD in **Quantum Molecular Science**, SOKENDAI, Japan
- Research fellow at **the University of Michigan**, USA
- Research fellow at **UC Berkeley**, USA
- Leader of **QuantumLab@VNUHCM-US**
- PI of NAFOSTED's project for **Strong Research Group in Quantum**
- Invited speaker at the Congress of **International Academy of Quantum Molecular Science (IAQMS)**, 2026

QuantumLab@VNUHCM-US

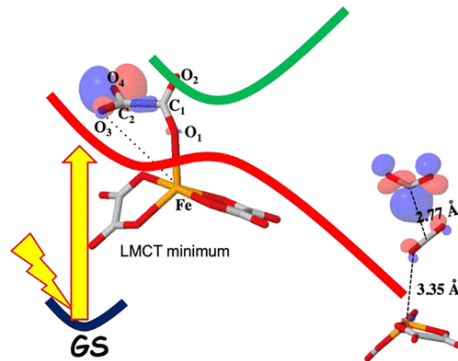
Electronic-structure methods

$$E = \langle 0 | e^{\hat{A}_{\text{int}}^\dagger} e^{\hat{A}_{\text{ext}}^\dagger} \hat{H} e^{\hat{A}_{\text{ext}}} e^{\hat{A}_{\text{int}}} | 0 \rangle$$

$$= \langle 0 | e^{\hat{A}_{\text{int}}^\dagger} \hat{H}_{\text{ext}} e^{\hat{A}_{\text{int}}} | 0 \rangle$$



Excited states and Dynamics



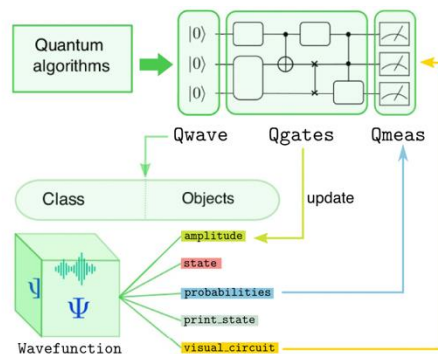
In collaboration with:

- Prof. Phan N. Loan, HCMUE
- Dr. Huynh T. Duc, VAST
- Dr. Le B. Ho, Tohoku, Japan
- Dr. Vu Tuan Hai, UIT
- Dr. Nguyen T. Hoang, VAST
- Quoc-Chuong Nguyen, DTU

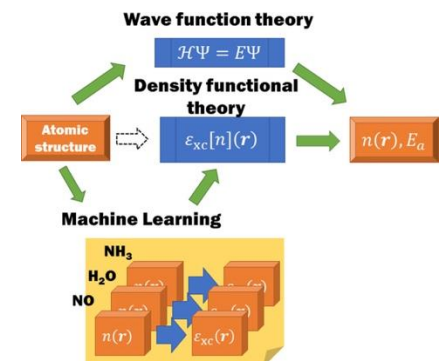
25 members



Quantum computing and information



Machine learning for many-body physics



Quantum Technology



Quantum Computing



Communication



Quantum Sensing

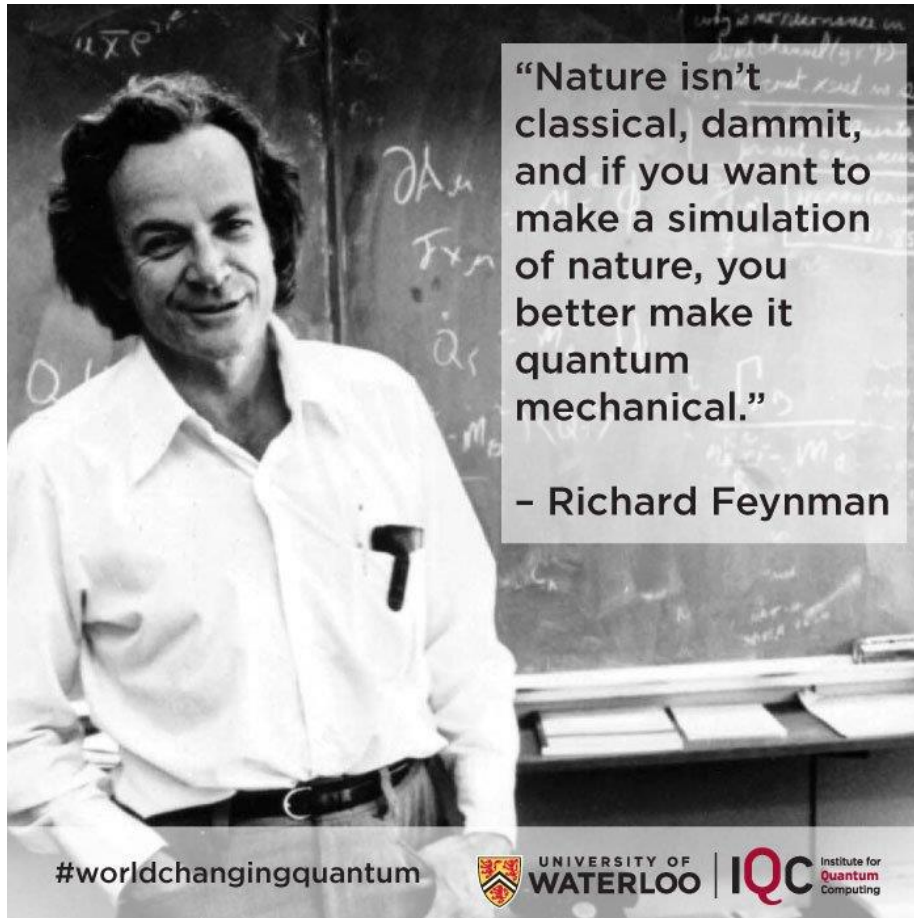


Simulation

Outline

1. What is Quantum computing (QC)?
2. Quantum computing program at VNUHCM-US
3. Quantum research in QuantumLab

1. What is quantum computing?



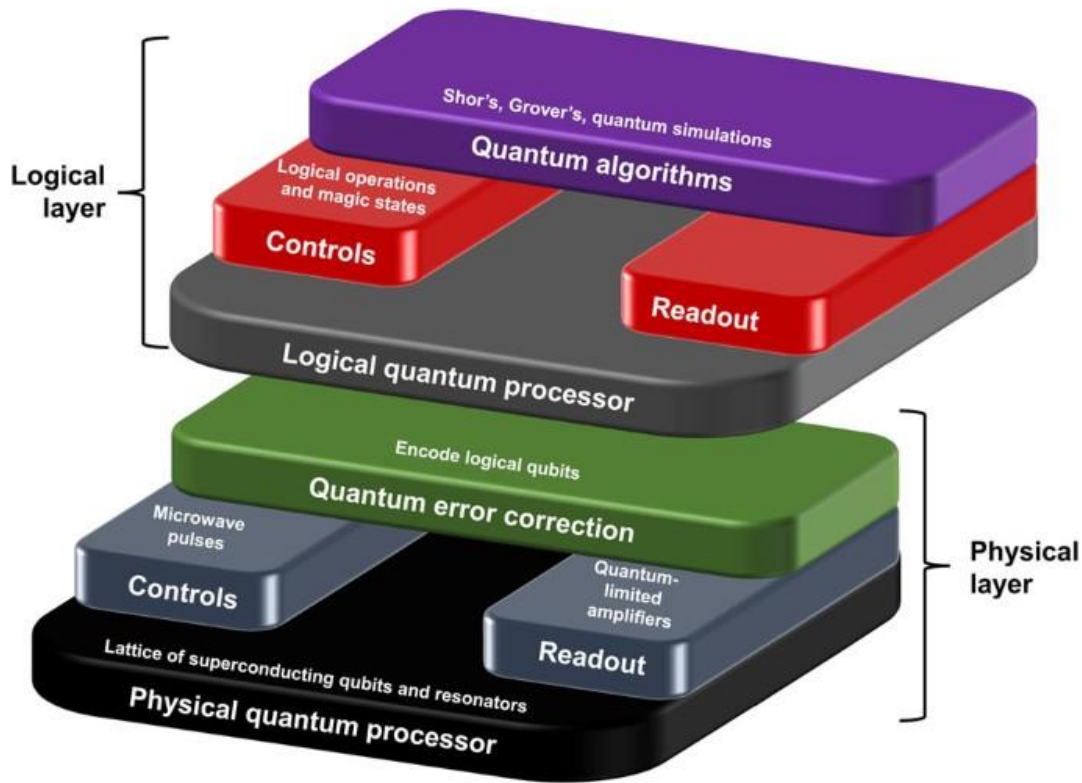
Quantum computing is a multidisciplinary field comprising aspects of computer science, physics, and mathematics **that utilizes quantum mechanics to solve complex problems faster than on classical computers.**



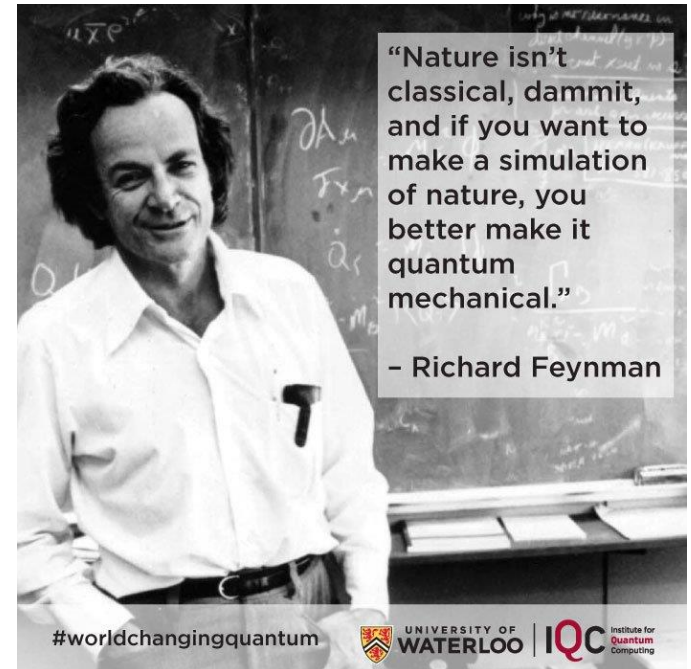
“Quantum computer Jiuzhang 2.0 is **10^{24} times faster than** current supercomputer.”

Yulin Wu *et al.* Phys. Rev. Lett. **2021**

1. What is quantum computing?



J. M. Gambetta et al. Nature Communications (2017).



Quantum computing is a multidisciplinary field comprising aspects of computer science, physics, and mathematics **that utilizes quantum mechanics to solve complex problems faster than on classical computers.**

1. What is quantum computing?

Qubits

N qubits can represent 2^N states

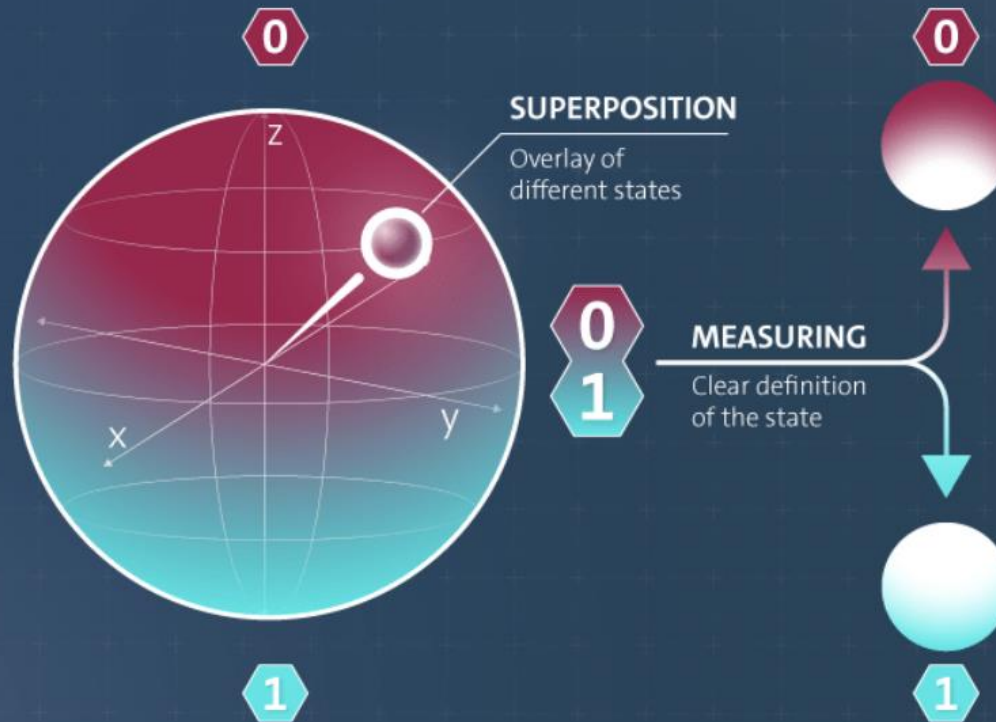
Classical Bit

Binary system



quantum bit “qubit”

Arbitrarily manipulable two-state quantum system



Parallel arithmetic operations possible

Exponential multiplication per qubit

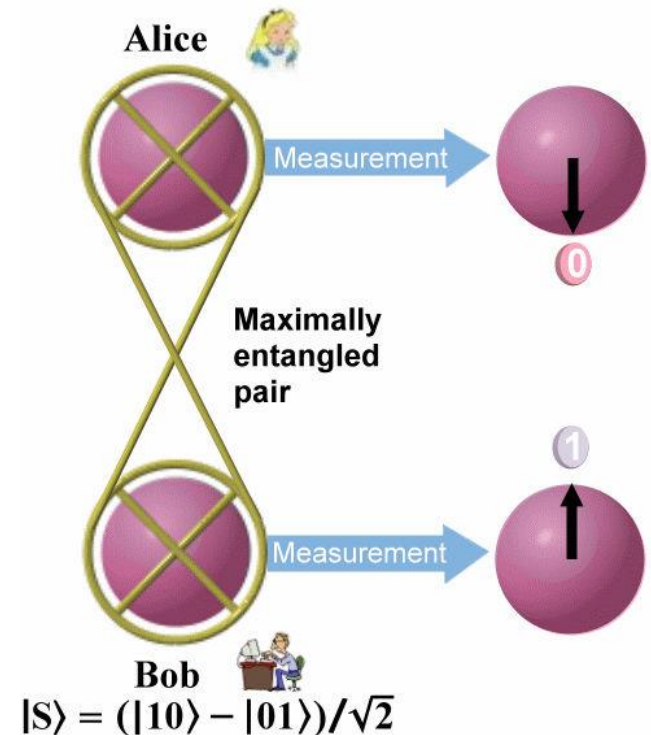
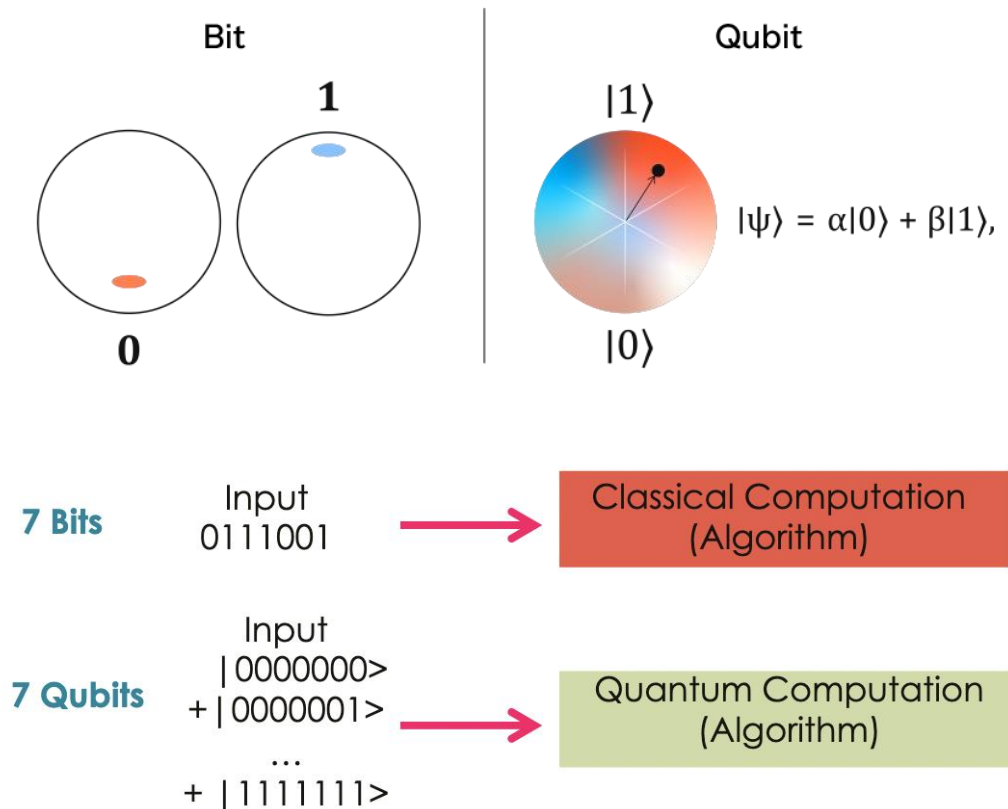
Massive amounts of data can be handled in plausible time

Taken from IBM

1. What is quantum computing?

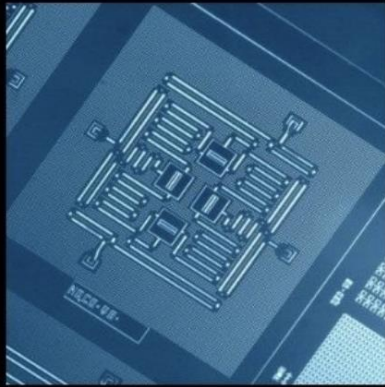
Qubits

- **Superposition:** A qubit can exist in both 0 and 1 states simultaneously
- **Entanglement:** Two or more qubits can be linked together regardless of distance.

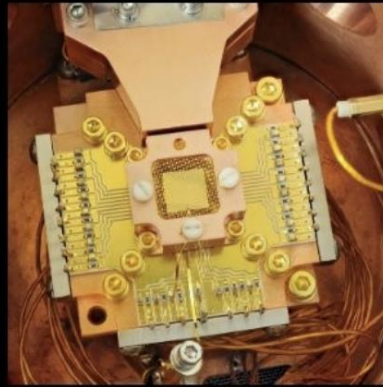


1. What is quantum computing?

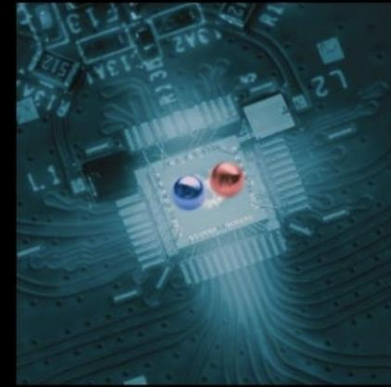
Types of quantum computers



SUPERCONDUCTING



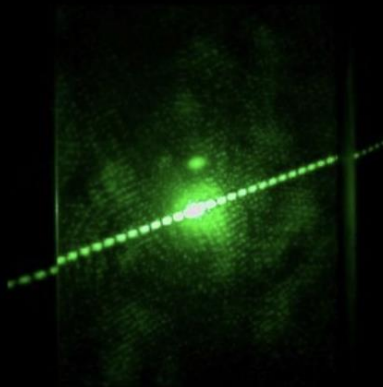
TRAPPED ION



SPIN



TOPOLOGICAL



PHOTONIC



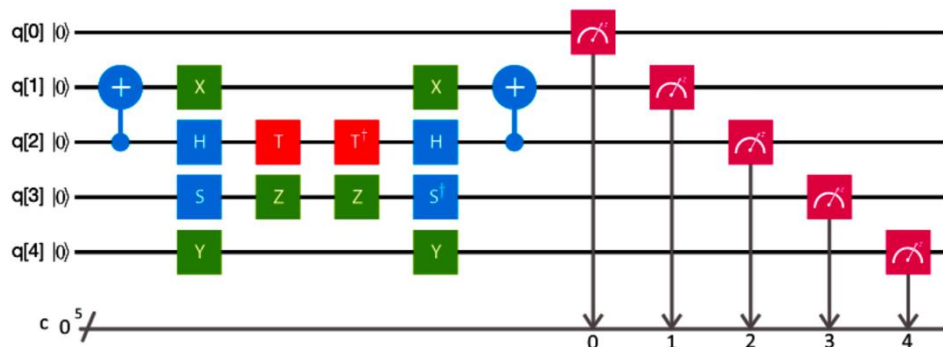
NEUTRAL ATOM

1. What is quantum computing?

Quantum circuits

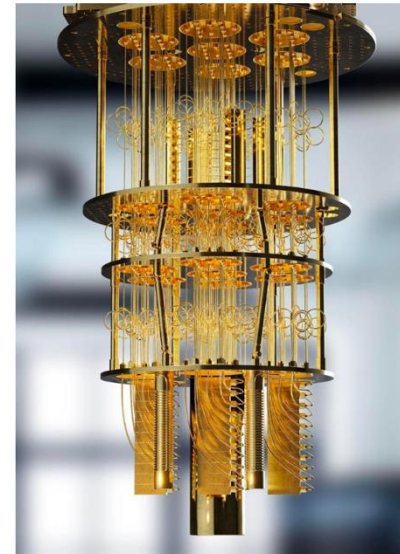
A quantum circuit is the quantum computing equivalent of a classical digital circuit.

A quantum circuit represents a universal set of quantum gates and qubits that perform specific computations.



includes:

- a set of qubits
- a set of quantum gates
- measurements

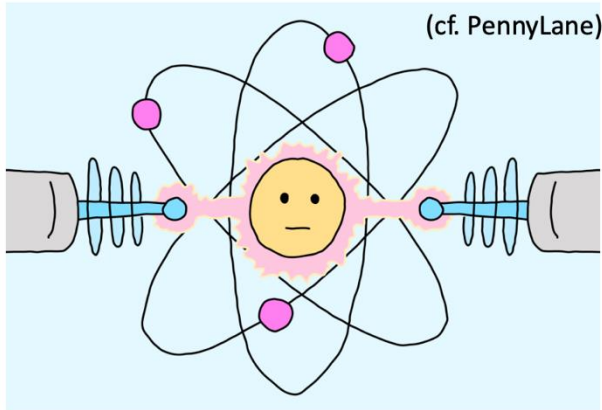


ibm quantum computer

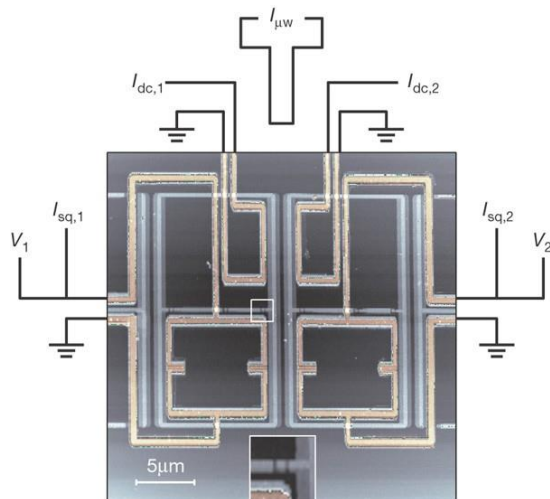
1. What is quantum computing?

Quantum gates

unitary matrices



.... operators that manipulate one or more qubits to transform the quantum state...



X Gate
Bit-flip, Not

$$\begin{bmatrix} X \end{bmatrix} \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \beta|0\rangle + \alpha|1\rangle$$

Z Gate
Phase-flip

$$\begin{bmatrix} Z \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

H Gate
Hadamard

$$\begin{bmatrix} H \end{bmatrix} \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{\alpha + \beta|0\rangle + \alpha - \beta|1\rangle}{\sqrt{2}}$$

T Gate

$$\begin{bmatrix} T \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha|0\rangle + e^{i\pi/4}\beta|1\rangle$$

Controlled Not
Controlled X
CNot

$$\begin{bmatrix} X \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = a|00\rangle + b|01\rangle + d|10\rangle + c|11\rangle$$

Swap

$$\begin{bmatrix} \text{Swap} \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = a|00\rangle + c|01\rangle + b|10\rangle + d|11\rangle$$

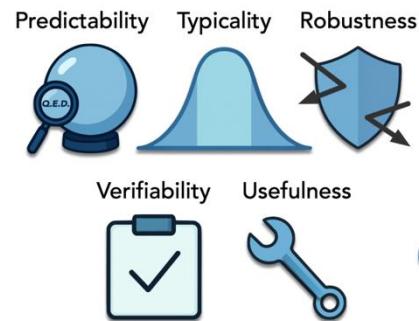
1. What is quantum computing?

- Quantum advantage is a demonstrated capability of quantum computers to perform a specific task with **superior efficiency, accuracy, or cost-effectiveness** compared to the best classical methods, often by leveraging the unique properties of quantum mechanics to solve problems currently intractable for even the most powerful supercomputers.

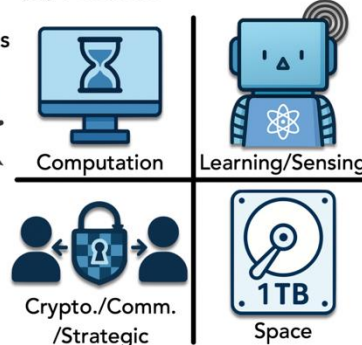


John P. Preskill, CALTECH

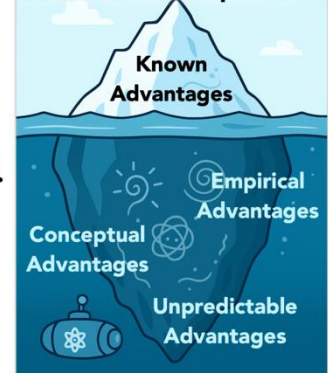
(a) Keystone Properties



(b) Realms



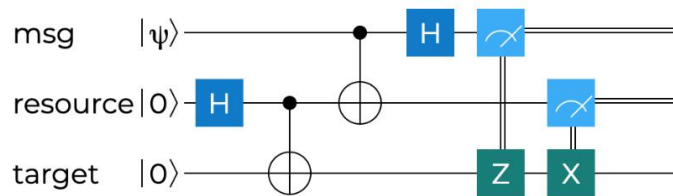
(c) Future Prospects



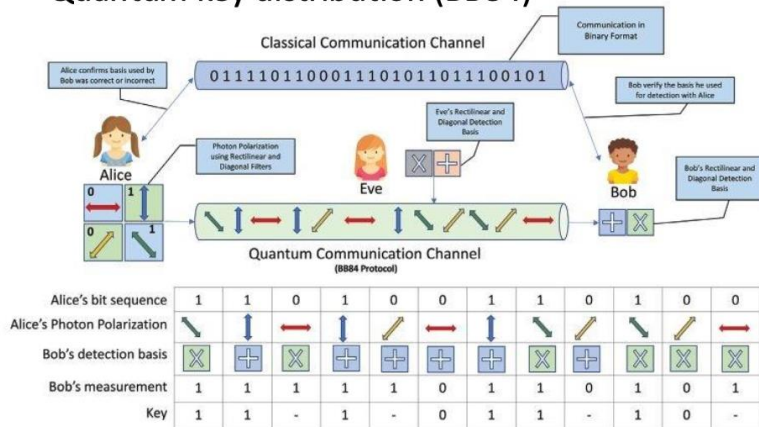
1. What is quantum computing?

Quantum Speedup

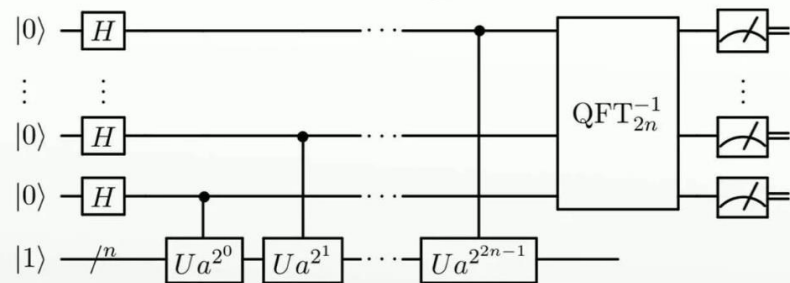
Quantum teleportation



Quantum key distribution (BB84)

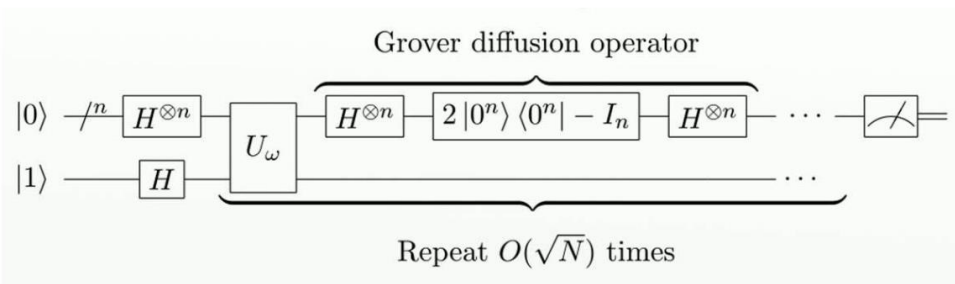


Shor's Algorithm (factoring)

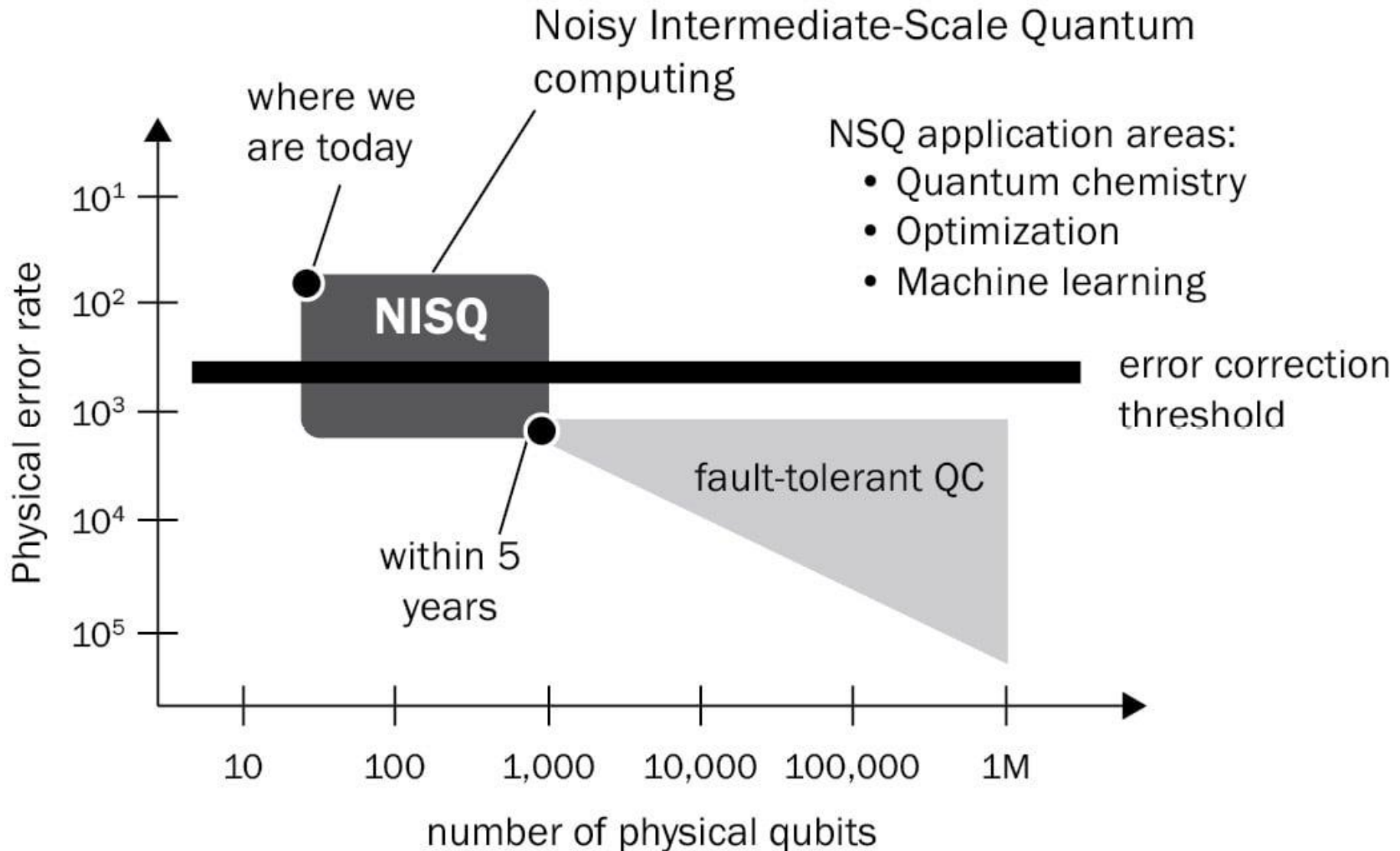


https://en.wikipedia.org/wiki/File:Shor's_algorithm.svg

Grover's Algorithm (search)



1. What is quantum computing?



1. What is quantum computing?

nature communications



Article

<https://doi.org/10.1038/s41467-023-37587-6>

Evaluating the evidence for exponential quantum advantage in ground-state quantum chemistry

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Check for updates

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CAN QUANTUM COMPUTING CRACK THE BIGGEST CHALLENGES IN HEALTH?

Quantum computing can model molecular interactions for drug discovery and analyze complex health datasets, but better hardware is needed. **By Marianne Guenot**

Cleveland Clinic in the USA and the UK National Health Service are both part of a burgeoning community of researchers, health systems and companies exploring applications of quantum computing in health. Although this nascent technology is at least a few years away from having practical

applications – and some are warning of the impending burst of the ‘quantum bubble’ – many experts believe that biomedical and healthcare research needs to be ready for the quantum revolution of tomorrow.

“I know we are taking a gamble,” says Lara Jehi, Cleveland Clinic’s chief research information officer, who leads the institute’s

quantum research. Jehi hopes that “there is a strategic, long-term advantage in being the group that was first.”

Health applications

Since last year, those grabbing food at the Cleveland Clinic medical research center cafeteria have had an odd companion at the

1. What is quantum computing?

Limitations and Challenges



Mitigate Noise and Decoherence

Focus on developing techniques to reduce environmental interference and maintain quantum state integrity.

Improve Error Correction

Invest in efficient error correction methods to ensure data integrity with minimal resource usage.

Enhance Scalability

Research and develop larger, more efficient quantum systems to handle complex computations effectively.

Hope but not hype!

1. What is quantum computing?

QUANTUM COMPUTERS IN DEVELOPMENT



1

GOOGLE is spending billions of dollars to build its quantum computer by 2029. The company opened a campus in California called Google AI to help it meet this goal.

2

IBM plans to have a 1,000-qubit quantum computer in place by 2023. For now, IBM allows access to its machines for those research organizations.

3

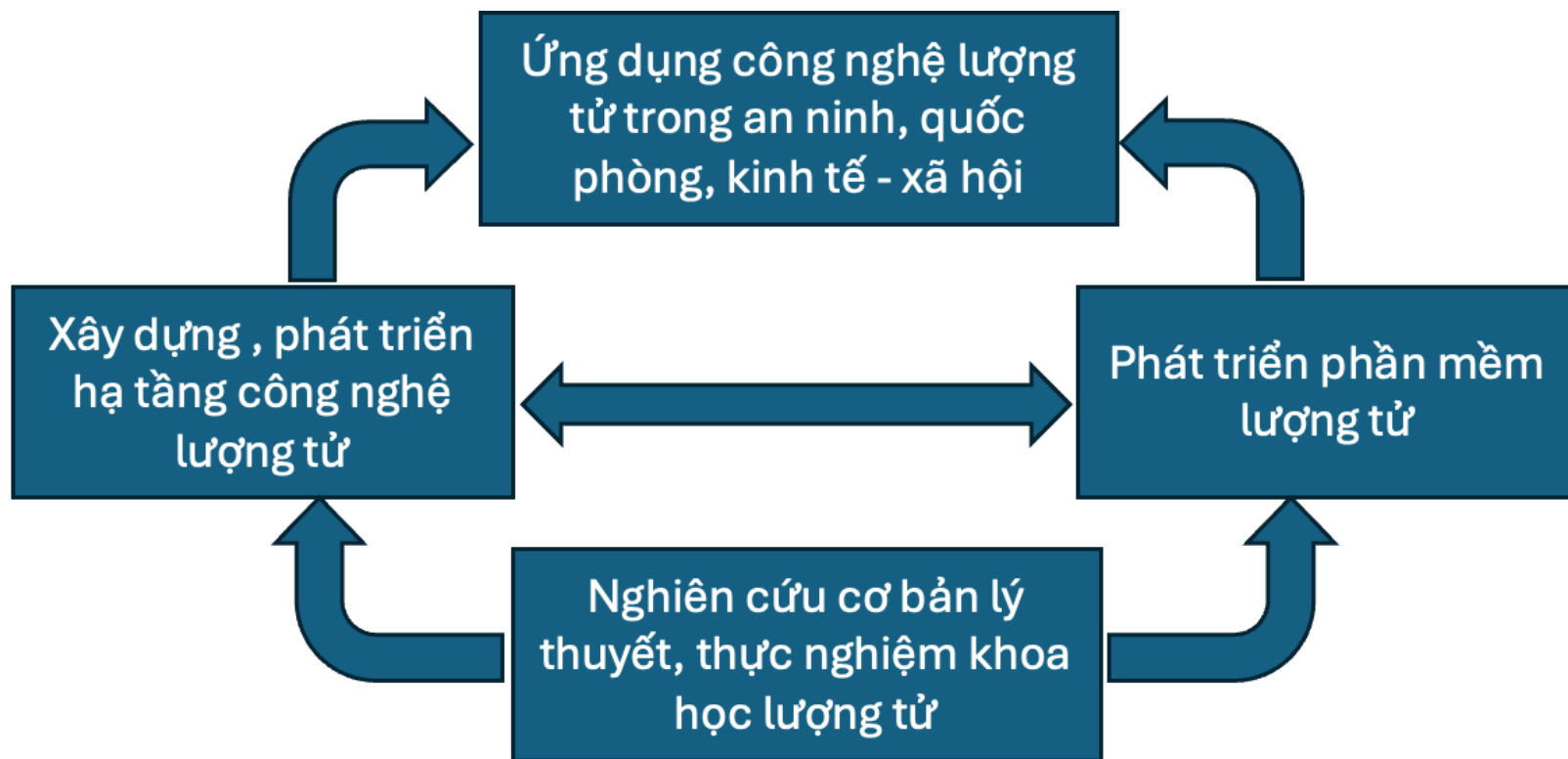
MICROSOFT offers companies access to quantum technology via the Azure Quantum platform.

4

OTHERS There's interest in quantum computing and its technology from financial services firms such as JPMorgan Chase and Visa.

1. What is quantum computing?

1. Nghị quyết 57-NQ/TW, ngày 22/12/2024 của Bộ Chính trị
2. Quyết định 1131/QĐ-TTg, ngày 12/6/2025 của Thủ tướng Chính phủ



Outline

1. What is Quantum computing (QC)?
2. Quantum computing program at VNUHCM-US
3. Quantum research in QuantumLab

2. QC program at VNUHCM-US

- Nền tảng vật lý: đại cương và lượng tử
- Nền tảng toán học: đại số tuyến tính, xác suất thống kê
- Nền tảng lập trình và tư duy thuật toán
- Học thuật toán cơ bản và nâng cao
- Thực hành lập trình tính toán lượng tử (trên máy lượng tử ảo)
- Hướng đến thiết kế thuật toán và giải quyết các bài toán thực tế.

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2. QC program at VNUHCM-US

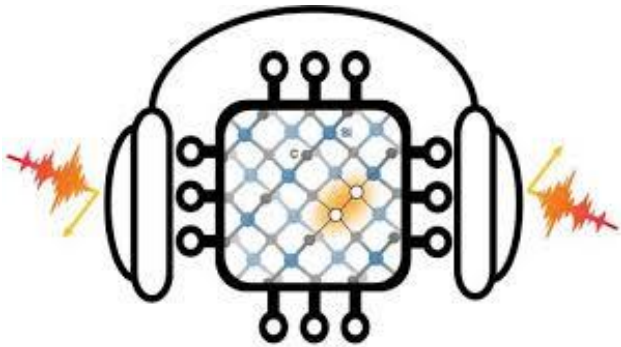
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Outline

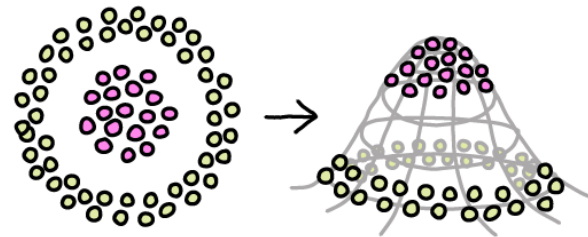
1. What is Quantum computing (QC)?
2. Quantum computing program at VNUHCM-US
3. Quantum many-body simulation

Our group is working on ...

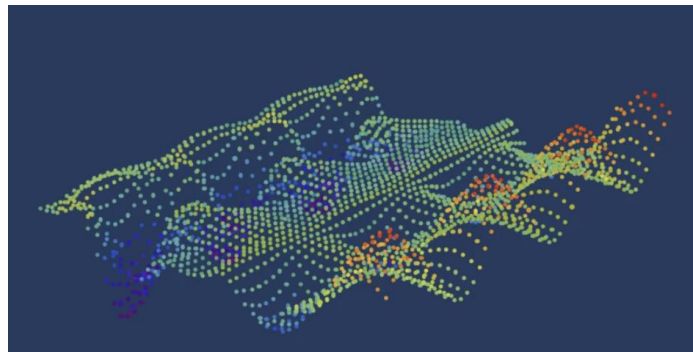
**Quantum many-body
simulation**



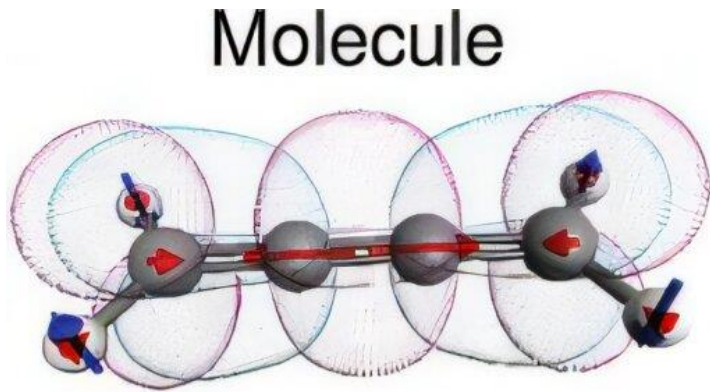
**Quantum machine
learning (QML)**



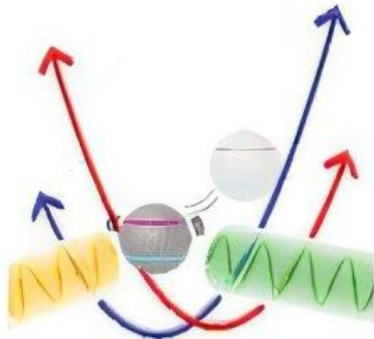
Quantum optimization



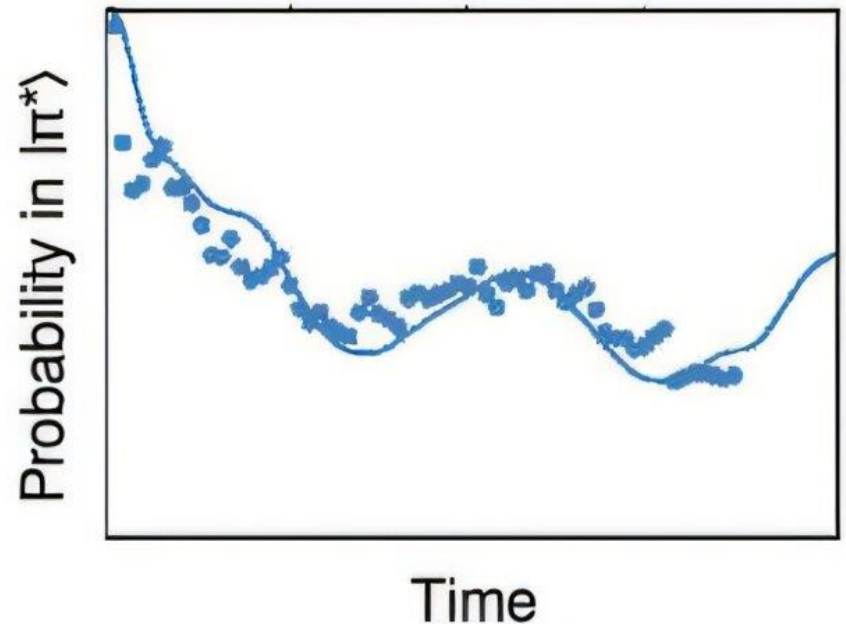
3. Quantum many-body simulation



Quantum simulator



$$i\hbar \frac{\partial \psi}{\partial t} = \underbrace{-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}}_{\text{K.E.}} + \underbrace{V \cdot \psi}_{\text{P.E.}}$$

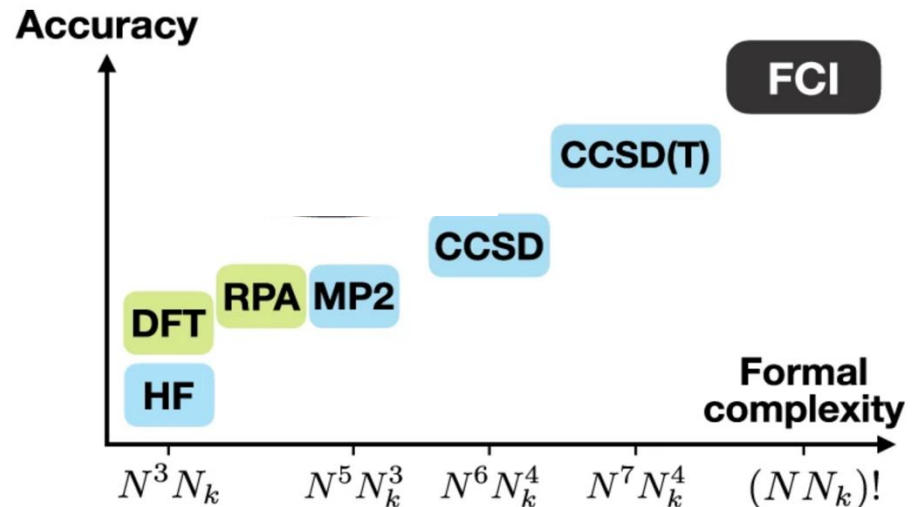
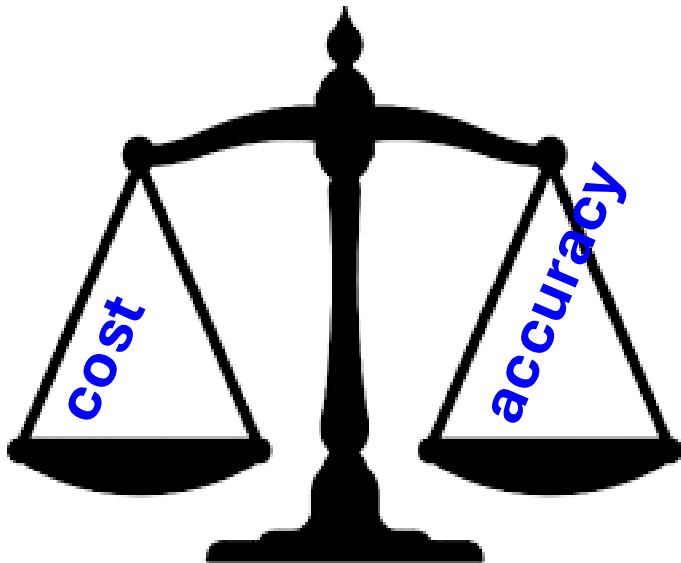


3. Quantum many-body simulation

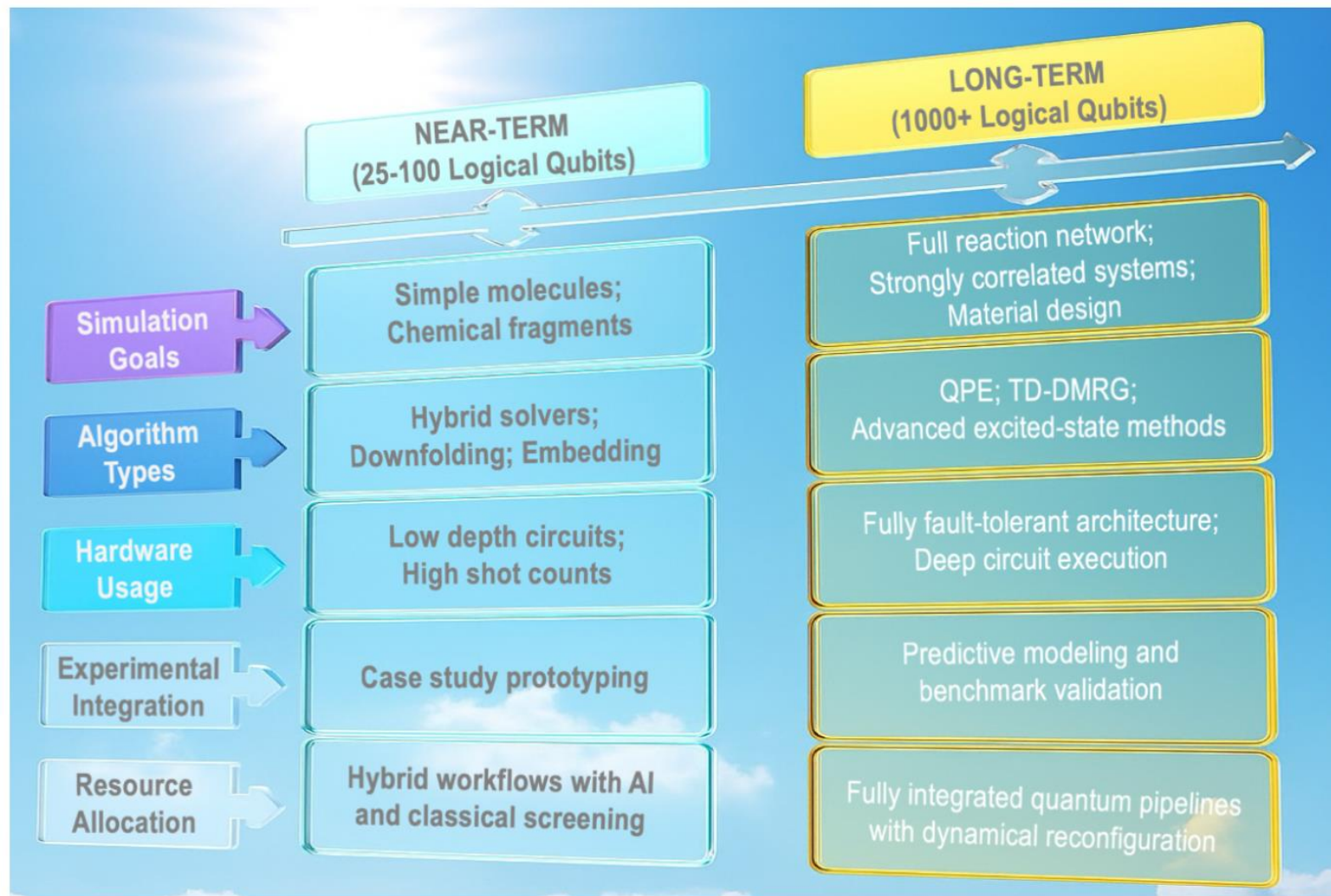
- In the Born-Oppenheimer approximation, molecular electronic Hamiltonian has form:

$$H_e = -\sum_i \frac{\nabla_i^2}{2} - \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}.$$

- Solving the Schrodinger equation becomes intractable when the number of degrees increases.



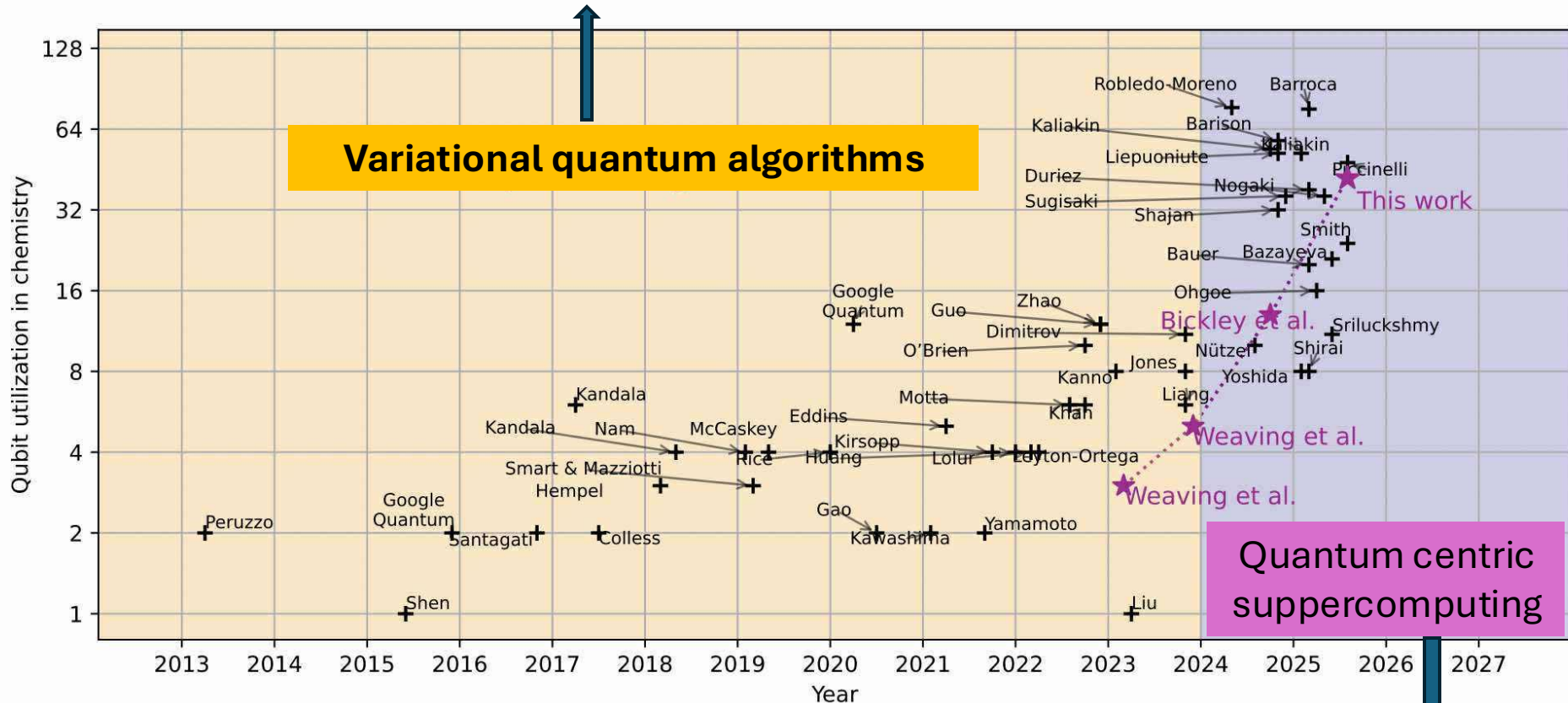
3. Quantum many-body simulation



3. Quantum many-body simulation

Quantum computing solvers

- Quantum computers measure cost functions
- Classical computers optimize cost functions



- Quantum computers sample states' space
- Classical computers diagonalize Hamiltonian matrix

3. Quantum many-body simulation

- **Variational quantum eigensolver (VQE)** is widely-used
- **VQE** is a classical-quantum hybrid algorithm
- **VQE** lies on the variational principle

$$E_0 \leq \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \quad \Rightarrow \quad E_{\text{VQE}} = \min_{\theta} \langle \mathbf{0} | U^\dagger(\theta) \hat{H} U(\theta) | \mathbf{0} \rangle$$



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Physics Reports

journal homepage: www.elsevier.com/locate/physrep



The Variational Quantum Eigensolver: A review of methods and best practices



Jules Tilly^{a,b,*}, Hongxiang Chen^{b,c}, Shuxiang Cao^{b,d}, Dario Picozzi^{a,b},
Kanav Setia^e, Ying Li^f, Edward Grant^{b,c}, Leonard Wossnig^{b,c}, Ivan Rungger^g,
George H. Booth^h, Jonathan Tennyson^a

3. Quantum many-body simulation

VQE: general scheme

Fermionic Hamiltonian problem

Hartree-Fock solution $|\psi_{HF}\rangle$



Problem mapped to qubits

$$H_{el} = \sum_j \alpha_j P_j$$

$$P_j = \prod_i \sigma_i^j, \sigma_i^j \in \{I, X, Y, Z\}$$



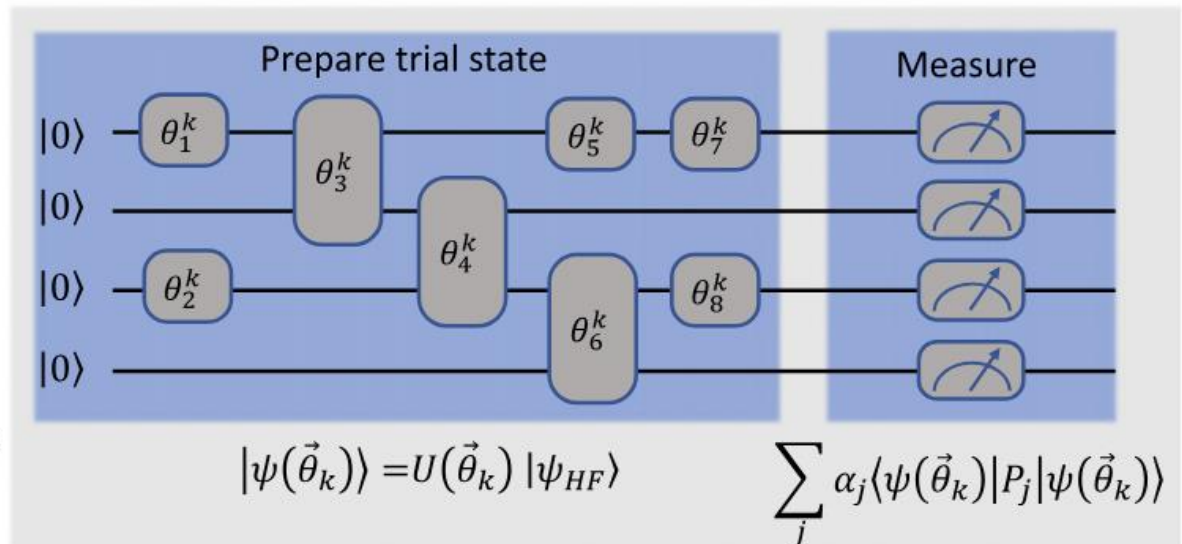
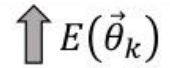
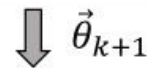
Generate ansatz with $\vec{\theta}_0$

$$|\psi(\vec{\theta}_0)\rangle = U(\vec{\theta}_0) |\psi_{HF}\rangle$$



Classical optimization algorithm:

$$\vec{\theta}_k \rightarrow \vec{\theta}_{k+1}$$



3. Quantum many-body simulation

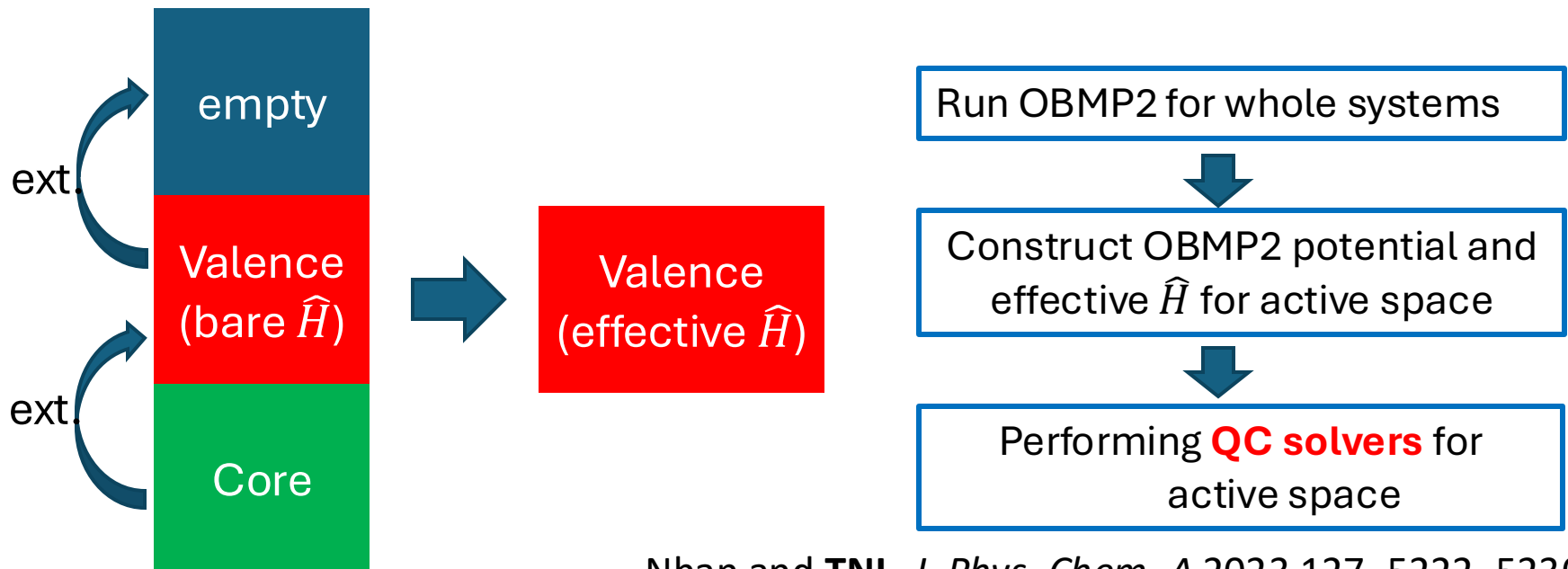
Space reduction: quantum downfolding

$$E = \langle \mathbf{0} | e^{\hat{A}_{\text{int}}^\dagger} e^{\hat{A}_{\text{ext}}^\dagger} \hat{H} e^{\hat{A}_{\text{ext}}} e^{\hat{A}_{\text{int}}} | \mathbf{0} \rangle$$

$$\hat{H}_{\text{ext}} = \hat{H} + \hat{v}_{\text{OBMP2}}$$

$$= \langle \mathbf{0} | e^{\hat{A}_{\text{int}}^\dagger} \hat{H}_{\text{ext}} e^{\hat{A}_{\text{int}}} | \mathbf{0} \rangle$$

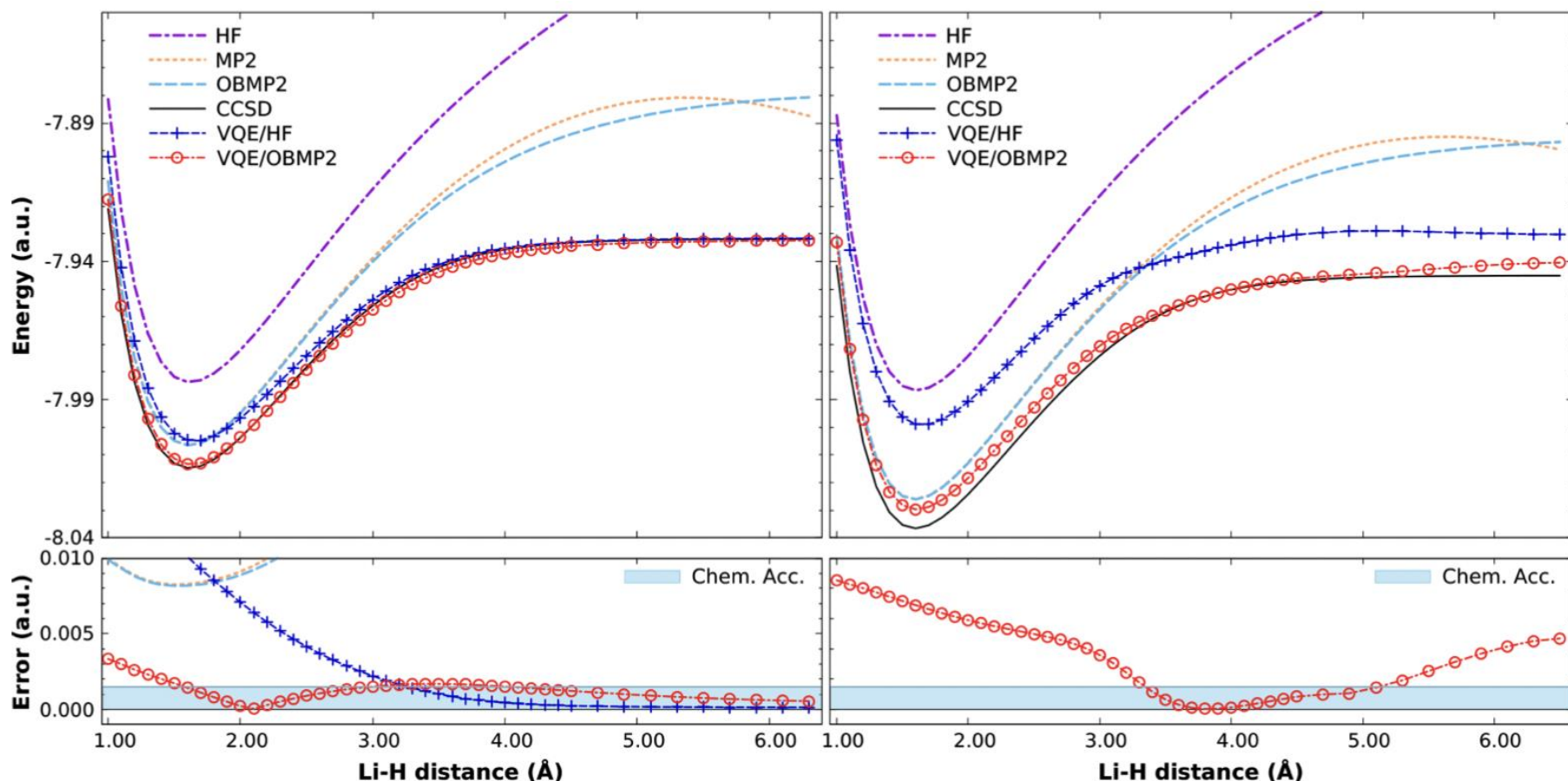
- Active space approximation for effective $\hat{H}$
- **Solving using VQE**



3. Quantum many-body simulation

VQE with quantum downfolding

LiH dissociation



3. Quantum many-body simulation

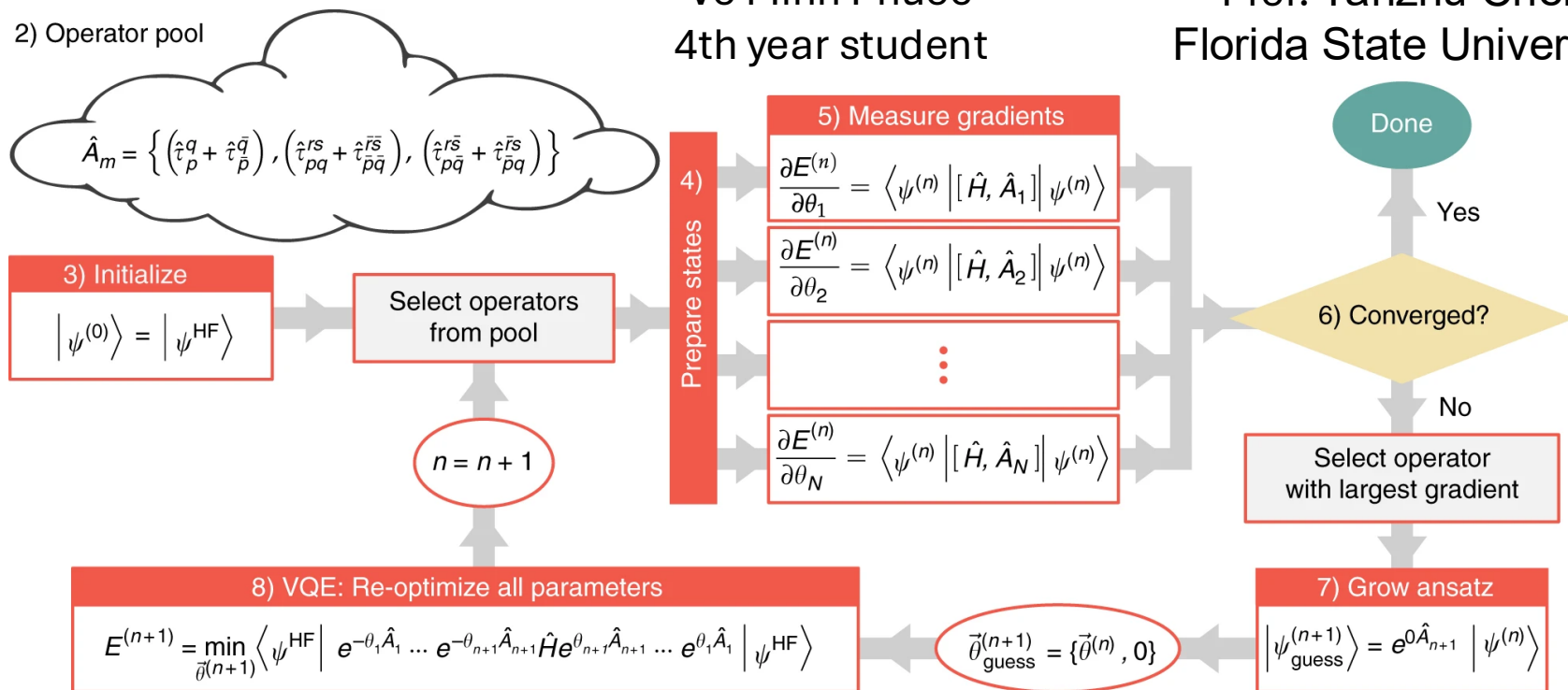
Adaptive VQE



Vo Minh Phuoc
4th year student

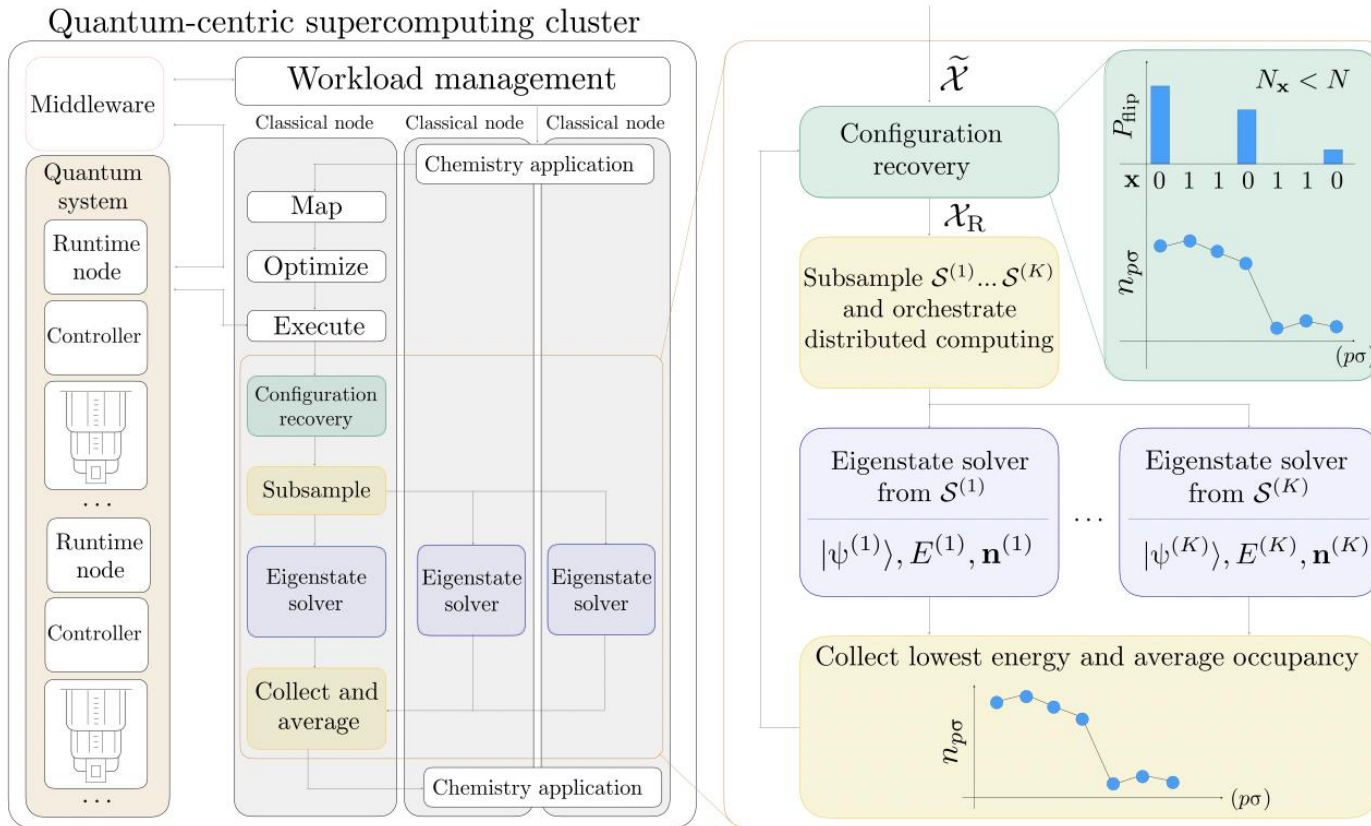


Prof. Yanzhu Chen
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3. Quantum many-body simulation

Quantum-centric supercomputing



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Thank you very much for your attention!



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